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Christian College of Engineering
ACC No..
BHIL AI

# 322612 (22)

BE (6th Semester)

Examination, April-May, 2013

Branch : CSE

## COMPILER DESIGN

*Time Allowed : Three Hours*

*Maximum Marks : 80*

*Minimum Pass Marks : 28*

**Note :** (i) All questions carry equal marks.

(ii) Sub question (a) is compulsory. Attempt any TWO subparts from (b), (c) & (d).

**Q. 1.** (a) What is Cross Compiler ? 2

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(b) Give examples of the errors that is encountered in lexical analyzer, syntax analyzer and semantic analyzer phases of compiler. 7

(c) What are the different compiler construction tools ? 7

(d) Explain Parser Construction with YACC ? 7

**Q. 2.** (a) Write the four component of context free grammar. 2



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(b) Define Left most and Right most derivation.

Also find the left most and right most derivation for the corresponding grammar

$S \rightarrow a|^{\wedge}|(T)$

$T \rightarrow T,S \mid S$

where string is (((a, a), ^, (a)), a) 7

(c) Construct SLR parsing for the following grammar 7

$S \rightarrow 0S0 \mid 1S1 \mid 10$



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(d) Explain the process of LR parsing with a suitable example. 7

Q. 3. (a) Define Syntax directed definition. 2

(b) Explain what is done in Syntax directed translation scheme ? Draw a figure showing conceptual view of syntax-directed Translation. 7

(c) (i) Define how synthesized and inherited attributes are used in Syntax directed definitions. 5



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(ii) Translate the following expression into postfix form.

(1) a*(b + c)

(2) (a + b)*(cc + d)

(d) Write short notes on : 7

(i) Three Address Codes

(ii) Translation of Assignment Statements

Q. 4. (a) Describe flow graphs. 2



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(b) Write the quadruple, triple and indirect triple for the assignment 7

X = (a + b)*- c/d

(c) Explain the various stages in the process of Run Time Environment. 7

(d) Explain the process of parameter passing in symbol table with suitable example. 7

Q. 5. (a) What do you mean by ambiguous grammar ? 2



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(b) Explain Code generator. What are the different problems are occurred in code generation ? 7

(c) Construct LALR parsing for the following grammar : 7

$S \rightarrow Aa \mid bAc \mid dc \mid bda$

$A \rightarrow d$

(d) Explain loop optimization. What are different steps for loop optimization ? 7



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